1. (a) Consider the causal, complex coefficient transfer function shown below:

$$H_M(z) = \frac{\sum_{n=0}^{n=M-1} d^*[M-n]z^{-n} + z^{-M}}{1 + \sum_{n=1}^{n=M} d[n]z^{-n}}, M > 1 \quad (1)$$

(i) Show that the transfer function in (1) is an allpass filter. **[5]**

Answer

The condition $|H_M(z)| = 1$, for $|z| = 1$ implies $\left|H_M\left(e^{j\omega}\right)\right| = 1$ for all ω. That means we are looking for an allpass transfer function. An Mth order causal, real-coefficient all pass transfer function must be of the form

$$H_M(z) = \frac{\sum_{n=0}^{n=M} d^*[M-n]z^{-n}}{\sum_{n=0}^{n=M} d[n]z^{-n}}$$

If we denote the denominator polynomial of the allpass function as $A_M(z)$ with

$$A_M(z) = \sum_{n=0}^{n=M} d[n]z^{-n}$$

then it follows that $H_M(z)$ can be written as:

$$H_M(z) = \frac{z^{-M}A_M(z^{-1})}{A_M(z)}$$

From the above we see that:

$$H_M(z^{-1}) = \frac{z^{M}A_M(z)}{A_M(z^{-1})}$$

$$H_M(z)H_M(z^{-1}) = \frac{z^{-M}A_M(z^{-1})}{A_M(z)}\frac{z^{M}A_M(z)}{A_M(z^{-1})} = 1 \Rightarrow \left|H_M\left(e^{j\omega}\right)\right|^2 = H_M(z)H_M(z^{-1}) = 1$$

$$\Rightarrow \left|H_M\left(e^{j\omega}\right)\right| = 1$$

Therefore, the condition $b[n] = a[M-n]$ must hold.

(ii) Comment on the locations of the zeros and the poles of a causal, real, stable, allpass filter's transfer function within the $z-$plane, with respect to the unit circle. Justify your answer. **[5]**

Answer

The poles of a causal stable transfer function must lie inside the unit circle. As a result, all zeros of a causal stable allpass transfer function lie outside the unit circle in a mirror-image symmetry with its poles situated inside the unit circle.

(b) Given an allpass filter

$$H(z) = \frac{d + ez + fz^2}{1 + bz + cz^2}$$

with poles at $\frac{1}{2}$ and $\frac{1}{3}$, find b, c, d, e, f. **[5]**

Answer

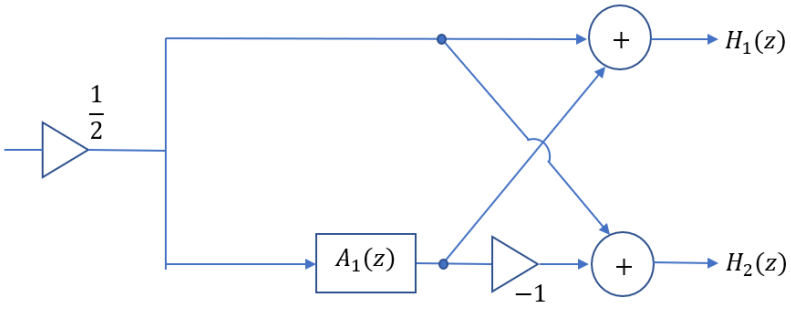
$$1 + bz + cz^2 = c\left(\frac{1}{c} + \frac{b}{c}z + z^2\right) = c\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right) = c\left(\frac{1}{6} - \frac{5}{6}z + z^2\right)$$

$$c = 6, b = -5$$

$$H(z) = \frac{d + ez + fz^2}{1 - 5z + 6z^2}$$

and therefore, $d = 6, e = -5, f = 1$.

(c) In the structure below, $A_1(z)$ is a first-order allpass transfer function. Show that $H_1(z)$ is a lowpass filter and $H_2(z)$ is a highpass filter.



[5]

Answer

$H_1(z) = \frac{1}{2}(1 + A_1(z))$
$H_1(z) = \frac{1}{2}(1 + \frac{-az+1}{z-a}) = \frac{1}{2}\frac{(1-a)(z+1)}{z-a}$
$z = 1, H_1(1) = 1$
$z = -1, H_1(-1) = 0$
Therefore, $H_1(z)$ is a lowpass filter.
$H_2(z) = \frac{1}{2}(1 - \frac{-az+1}{z-a}) = \frac{1}{2}\frac{(1+a)(z-1)}{z-a}$
$z = 1, H_1(1) = 0$
$z = -1, H_1(-1) = 1$
Therefore, $H_2(z)$ is a highpass filter

(d) Consider the cascade of two causal LTI systems: $h_1[n] = a\delta[n] + b\delta[n-1]$ and $h_2[n] = c^n u[n], |\gamma| < 1$. Determine the frequency response $H(e^{j\omega})$ of the overall system. Find all the possible sets of values of a, b, c for which the overall transfer function has an amplitude equal to 1 for all frequencies. **[5]**

Answer

$$H(z) = H_1(z)H_2(z) = \frac{(a + bz^{-1})z}{z - c} = \frac{az + b}{z - c}$$
$|H(e^{j\omega})| = 1$ implies that $H(z)$ is an allpass filter.
$$|H(e^{j\omega})| = \frac{a^2 + b^2 + 2ab\cos(\omega)}{1 + c^2 - 2c\cos(\omega)}$$

$a^2 + b^2 = 1 + c^2$
$ab = -c$
$a^2 + b^2 = 1 + a^2b^2$
$a^2(1 - b^2) = (1 - b^2)$

$a = 1 \Rightarrow b = -c$
$$H(z) = \frac{z + b}{z + b} = 1$$

$a = -1 \Rightarrow b = c$

$$H(z) = \frac{-z + b}{z - b} = -1$$

$b = 1 \Rightarrow a = -c$

$$H(z) = \frac{az + 1}{z + a}$$

$b = -1 \Rightarrow a = c$

$$H(z) = \frac{az - 1}{z - a}$$

2. (a) Design a length-5 FIR bandpass filter with an antisymmetric impulse response $h[n]$, satisfying the following magnitude response values: $\left|H(e^{j0.3\pi})\right| = 0.3$ and $\left|H(e^{j0.6\pi})\right| = 0.8$. **[4]**

Answer

$$H(e^{j\omega}) = h[0] + h[1]e^{-j\omega} - h[1]e^{-j3\omega} - h[0]e^{-j4\omega} = e^{-j2\omega}[h[0](e^{j2\omega} - e^{-j2\omega}) + h[1](e^{j\omega} - e^{-j\omega})]$$
$$= j2e^{-j2\omega}[h[0]\sin(2\omega) + h[1]\sin(\omega)]$$

$$\left|H(e^{j\omega})\right| = 2[h[0]\sin(2\omega) + h[1]\sin(\omega)]$$
Thus,
$$\left|H(e^{j0.3})\right| = 2[h[0]\sin(0.6) + h[1]\sin(0.3)] = 0.3$$
$$\left|H(e^{j0.6})\right| = 2[h[0]\sin(1.2) + h[1]\sin(0.6)] = 0.8$$
Solving these two equations we get
$h[0] = 2.5686$ and $h[1] = -0.9873$

(b) We have shown that a real-coefficient FIR transfer function $H(z)$ with a symmetric impulse response has a linear phase response. As a result, the all-pole IIR transfer function $G(z) = \frac{1}{H(z)}$ will also have a linear-phase response. What are the practical difficulties in implementing $G(z)$? Justify your answer mathematically. **[4]**

Answer

$G(z)$ must have poles outside the unit circle making it unstable.

(c) A simple averaging filter is defined as
$$y[n] = \frac{1}{N}(x[n] + x[n-1] + \cdots + x[n-(N-1)])$$
This is clearly an FIR Filter.

(i) Let $N = 4$. Determine the transfer function, its zeros and poles. **[3]**

Answer

$$Y(z) = \frac{1}{N}[1 + z^{-1} + \cdots + z^{-(N-1)}]X(z) = H(z)X(z)$$
$$H(z) = \frac{1}{N}\frac{1 - z^{-N}}{1 - z^{-1}} = \frac{1}{N}\frac{1}{z^{N-1}}\frac{z^N - 1}{z - 1}$$

For $N = 4$ we have $H(z) = \frac{1}{4}\frac{1}{z^3}\frac{z^4-1}{z-1}$
$z^4 - 1 = 0 \Rightarrow z^4 = 1 \Rightarrow z^2 = \pm 1$
$z^2 = 1 \Rightarrow z = \pm 1$
$z^2 = -1 \Rightarrow z = \pm j$
The zero at 1 is cancelled with the pole at 1.
Therefore, we have zeros at $-1, \pm j$ and a triple pole at 0.

(ii) Determine a general form for zeros and poles for any N. **[3]**

Answer

$$H(z) = \frac{1}{N}\frac{1}{z^{N-1}}\frac{z^N - 1}{z - 1}$$

Zeros are found from $z^N - 1 = 0 \Rightarrow z^N = 1 \Rightarrow z = e^{j\frac{2\pi k}{N}}, k = 0, \dots, N-1$
Poles at $z = 0$ with multiplicity $N - 1$ and a pole at $z = 1$ which is cancelled by the zero at $z = 1$.

Therefore, zeros are at $z = e^{j\frac{2\pi k}{N}}, k = 1, \dots, N-1$.

(iii) By comparing $y[n]$ and $y[n-1]$ determine a recursive implementation, and furthermore, the transfer function, together with its zeros and poles of the recursive implementation. Looking at this example, can we say that "any" recursive filter is IIR? **[3]**

$$y[n-1] = \frac{1}{N}(x[n-1] + x[n-2] + \cdots + x[n-1-(N-1)])$$
$$y[n] = \frac{1}{N}(x[n] + x[n-1] + \cdots + x[n-(N-1)])$$
$$y[n] - y[n-1] = \frac{1}{N}(x[n] - x[n-1-(N-1)]) \Rightarrow$$
$$y[n] - y[n-1] = \frac{1}{N}(x[n] - x[n-N])$$

The answer to the question is NO since the filter under consideration is both recursive and FIR.

(d) An FIR digital filter has the transfer function

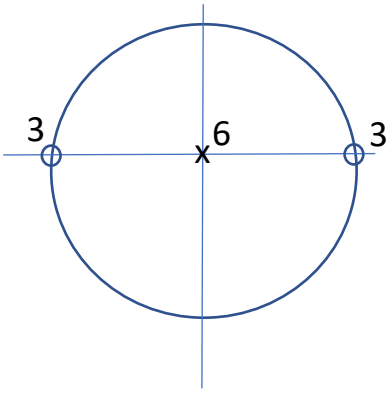
$$H(z) = (1 - z^{-1})^3(1 + z^{-1})^3$$

(i) Sketch the pole-zero diagram of this system. **[4]**

Answer

Using polynomial expansion, we obtain

$$H(z) = \frac{(z-1)^3(z+1)^3}{z^6}$$



(ii) Sketch roughly the amplitude response of the above filter. Would you classify this as a low-pass, high-pass, band-pass, or band-stop filter? Please briefly explain. **[4]**

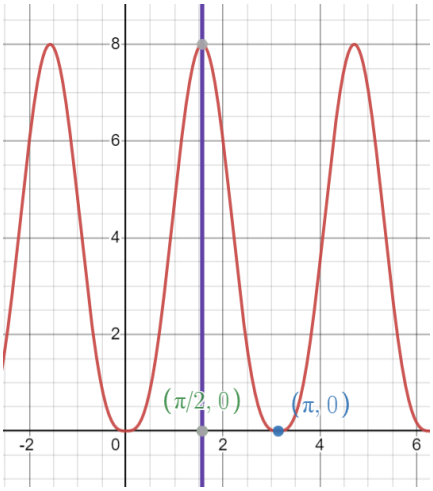
Answer

This is the Type 3 linear phase filter it can only be band pass.

$z = 1 \Rightarrow \omega = 0 \Rightarrow H(e^{j\omega}) = 0$

$z = -1 \Rightarrow \omega = \pi \Rightarrow H(e^{j\pi}) = 0$

$z = \pm j \Rightarrow \omega = \frac{\pi}{2} \Rightarrow H(e^{\pm j\pi/2}) = 8$



3. (a) The bilinear transformation from the s −plane to the z −plane is given by

$$s = \frac{1 - z^{-1}}{1 + z^{-1}}$$

(i) Explain using mathematical relations, where each point $s = \sigma + j\Omega$ on the s −plane is mapped on the z −plane through the bilinear transformation.

Answer

$z = \frac{1+s}{1-s}$

For $s = j\Omega_0$ we have that $z = \frac{1+j\Omega_0}{1-j\Omega_0}$ which has a magnitude equal to 1. This implies that a point on the imaginary axis in the s −plane is mapped onto a point on the unit circle in the z −plane where $|z| = 1$. In the general case, for $s = \sigma_0 + j\Omega_0$

$z = \frac{1+\sigma_0+j\Omega_0}{1-\sigma_0-j\Omega_0} \Rightarrow |z|^2 = \frac{(1+\sigma_0)^2 + {\Omega_0}^2}{(1-\sigma_0)^2 + {\Omega_0}^2}$

A point in the left half s −plane with $\sigma_0 < 0$ is mapped onto a point inside the unit circle in the z −plane as $|z| < 1$. Likewise, a point in the right half s −plane with $\sigma_0 > 0$ is mapped onto a point outside the unit circle in the z −plane as $|z| > 1$.

(ii) Prove that the relationship between the continuous-time angular frequency and the discrete-time angular frequency is non-linear.

Answer

We see that

$j\Omega = \frac{1-e^{-j\omega}}{1+e^{-j\omega}} = j\tan\left(\frac{\omega}{2}\right) \Rightarrow \Omega = \tan\left(\frac{\omega}{2}\right)$ which is a non-linear relationship.

(b) A given real lowpass digital IIR filter has a rational transfer function $H_L(z)$ and a cutoff frequency ω_c. The transfer function is transformed by replacing z by $F(z) = \frac{az+b}{cz+d}$, to a real highpass rational transfer function $H_H(z) = H_L(F(z))$ of the same order as $H_L(z)$ but different cutoff frequency. By using the constraints $H_H(e^{j\pi}) = H_L(e^{j0}) = 1$ and $H_H(e^{j0}) = H_L(e^{j\pi}) = 0$, derive relationships among the parameters a, b, c, d and explain what type of filter is $F(z)$.

Answer

$H_H(e^{j0}) = H_L\left(F(e^{j0})\right) = H_L(F(1)) = 0$

Therefore, $F(1) = \frac{a+b}{c+d} = e^{j\pi} = -1 \Rightarrow a + b = -c - d$

$H_H(e^{j\pi}) = H_L\left(F(e^{j\pi})\right) = H_L(F(-1)) = 1$

Therefore, $F(-1) = \frac{-a+b}{-c+d} = 1 \Rightarrow -a + b = -c + d$

Therefore, $b = -c$ and $a = -d$

$F(z) = \frac{az+b}{-bz-a} = -\frac{az+b}{bz+a}$

$F(z)$ is an allpass filter.

(c) Consider the Finite Impulse Response (FIR) filter transfer functions $G_1(z) = \frac{1}{4}(z + 2 + z^{-1})$ and $G_2(z) = \frac{1}{4}(-z + 2 - z^{-1})$. Explain the type of filters $G_1(z)$ and $G_2(z)$ (lowpass etc), by using the following two approaches:

(i) Intuitive approach. In this approach you can describe the effects of filters $G_1(z)$ and $G_2(z)$ on an input signal without necessarily using mathematical relationships. **[5]**

Answer

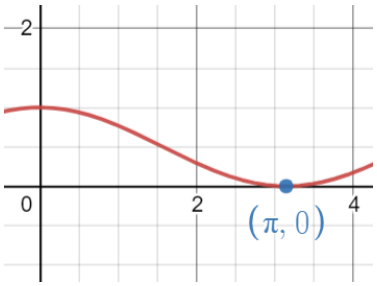
$G_1(z)$ is a lowpass filter because when we mix signal samples with positive weights we destroy the abrupt changes of the signals which are related to the high frequencies of the signal. $G_2(z)$ is a highpass filter because when we mix signal samples with some negative weights we might enhance the abrupt changes of the signals which are related to the high frequencies of the signal.

(ii) Mathematical approach. In this approach you must find the frequency response of the two filters. **[5]**

Answer

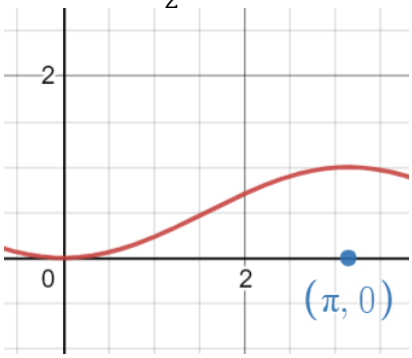
$G_1(z) = \frac{1}{4}(z + 2 + z^{-1}) = \frac{1}{4}(2 + 2\cos(\omega)) = \frac{1}{2}(1 + \cos(\omega))$

$|G_1(e^{j\omega})| = \frac{1}{2}|1 + \cos(\omega)|$. We observe that $|G_1(e^{j\omega})|$ decreases as the frequency approaches π.



$G_2(z) = \frac{1}{4}(-z + 2 - z^{-1}) = \frac{1}{4}(2 - 2\cos(\omega)) = \frac{1}{2}(1 - \cos(\omega))$

$|G_2(e^{j\omega})| = \frac{1}{2}|1 - \cos(\omega)|$



We observe that $|G_2(e^{j\omega})|$ increases as the frequency approaches π.

4. (a) Consider the multirate structure of **Figure 1** below, where $H_0(z), H_1(z)$ and $H_2(z)$ are ideal, zero phase, real-coefficient lowpass, bandpass and highpass filters respectively, with frequency responses as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{\pi}{3})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \frac{2\pi}{3})$$

$$H_2(e^{j\omega}) = u(\omega - \frac{2\pi}{3}) - u(\omega - \pi)$$

The function $u(\omega)$ is the well-known unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (-\frac{9\omega^2}{\pi^2} + 1)\left(u(\omega) - u\left(\omega - \frac{\pi}{3}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n]$ and $y_2[n]$.
Notice that all frequency domain representations above are provided only within the range $[0, \pi]$.

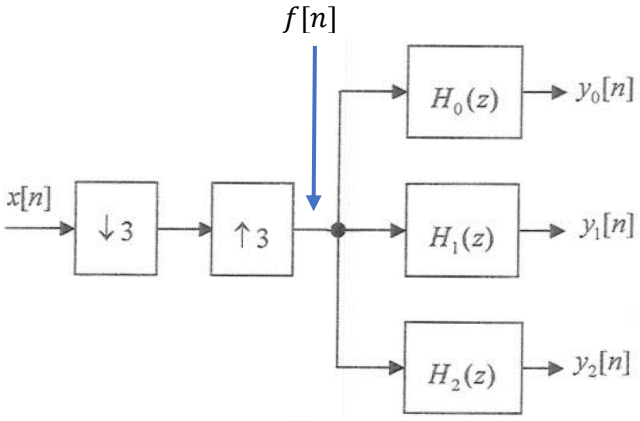
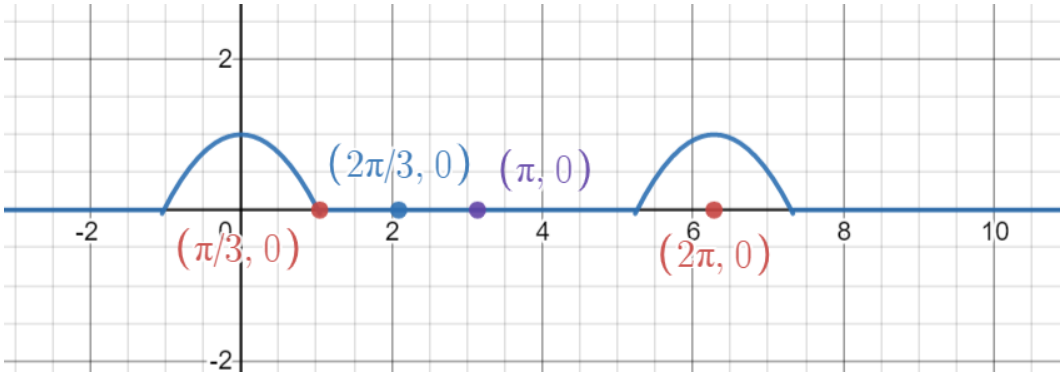


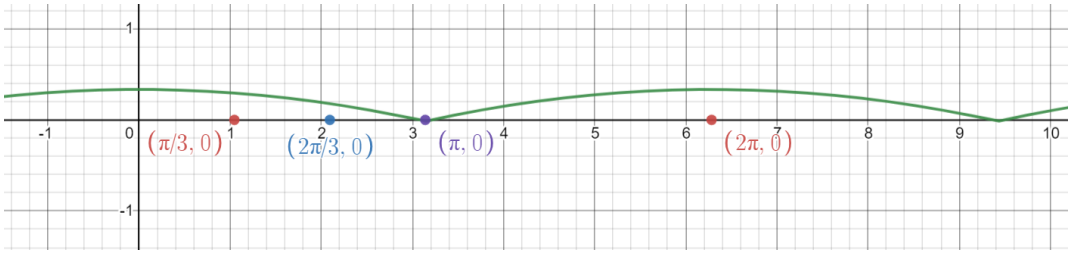
Figure 1

Answer

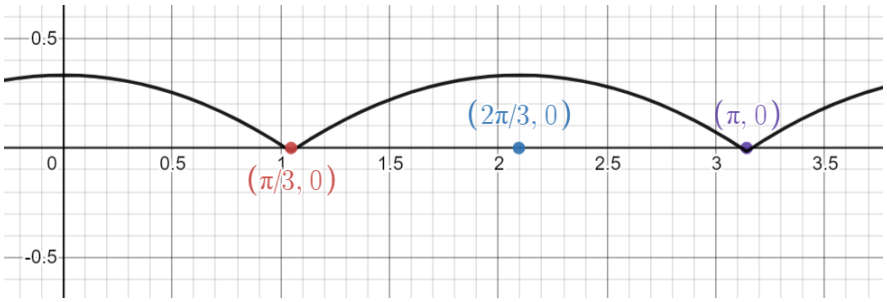
The DTFT of the input is shown below:



After decimation by 3:



After interpolation by 3 we obtain the DTFT of the signal $f[n]$ shown below.



The output of the three filters is obvious.

(b) Develop an expression for the output $y[n]$ as a function of the input $x[n]$ for the multirate structure of Figure 3 below.

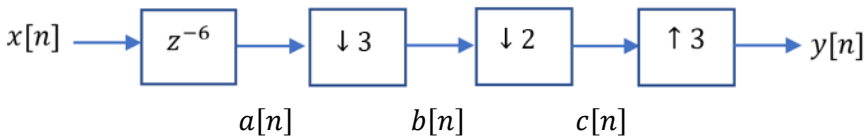


Figure 2

Answer

$a[n] = x[n-6]$
$b[n] = a[3n]$
$c[n] = b[2n] = a[6n] = x[6n-6]$
$y[n] = c\left[\frac{n}{3}\right] = x[2n-6]$